

JAMES WILLIAM “WILL” CARTER V

(210) 550-5033 | jcarte25@nd.edu | github.com/WillCarter5

EDUCATION

University of Notre Dame | Notre Dame, IN

Bachelor of Science

Major: Computer Science

May 2028

GPA: 3.9+

EXPERIENCE

Complex Networks Lab | Notre Dame, IN

Aug. 2025 - Present

Research Assistant

- Researches graph theoretical approaches to simulating protein folding and protein structure and function classification
- Designed and analyzed novel mathematical algorithm and developed a command-line tool for efficient protein structure comparison
- Validated a published study demonstrating an advanced network-based method for classifying protein 3D structures

University of Notre Dame Learning Resource Center | Notre Dame, IN

Aug. 2025 – Dec. 2025

Math Help Room Tutor

- Provided drop-in tutoring support for students in calculus, multivariable calculus, and linear algebra
- Led weekly small group lessons to reinforce students' understanding of linear algebra and differential equations

Southwest Research Institute | San Antonio, TX

Jun. 2023 - May 2024

Program Attendee, Research Intern

- Reprogrammed the PHIDRATES astrophysics website to a modern tech stack for improved maintainability

PROJECTS

Digital Integrated Circuits Course Projects

Aug. 2026 - Present

- Designed and verified standard cells in 0.5µm technology using Cadence Virtuoso, Layout GXL, and Analog Place and Route, achieving DRC/LVS compliance and successful parasitic extraction; synthesized and implemented an 8-bit MIPS processor using Genus and Innovus
- Designing a SystemVerilog processor for TSMC 180nm synthesis using Cadence Flowkit, with ATPG and signoff verification using Modus and Tempus, and power analysis using Voltus.

Full-Stack Web Application – WorNDly | Programming Paradigms

Jan. 2026 - Apr. 2026

- Developed and deployed a full-stack word-guessing web application using Python, Django, SQLite, and JavaScript, implementing user authentication, multilingual gameplay, input validation, persistent game history, and player statistics, as well as REST API integration for virtual currency transactions
- Collaborated through GitHub using version control and documented team contributions

Notre Dame Walking Directions | Data Structures and Algorithms

Dec. 2025

- Developed an optimal walking direction finder in Python using Dijkstra's weighted graph traversal algorithm and employing OpenStreetMap queries, matplotlib, and pandas
- Configured unit tests and timing analyses to examine the efficacy of the code

Personal Browser Home Page Dashboard

Sep. 2025

- Synthesized a full-stack browser dashboard with 20+ services, including real-time weather and NASA Astronomy Picture of the Day widgets
- Built the React/TailwindCSS frontend and managed a multi-route ExpressJS/Node.js backend with caching, scheduling, and throttling

LEADERSHIP AND ACTIVITIES

Boy Scouts of America | San Antonio, Texas

2015 - 2024

Eagle Scout

Residential Life in Alumni Hall | Notre Dame, Indiana

2026

Website Commissioner, Academic Commissioner

TECHNICAL AND LANGUAGE SKILLS

Technical: Git, Linux, Vim, Visual Studio Code, Android Studio, Cadence Virtuoso

Languages: C, C++, Python, Julia, Java, Bash, JavaScript, SQL, Django, React,

Coursework: Introduction to Artificial Intelligence, Programming Paradigms, Digital Integrated Circuits, Database Concepts, Operating Systems, Design and Analysis of Algorithms, Computer Networks